

Shift Invariance

1 Motivation

Our goal is to extend the QFT framework to encompass a broader class of gates, particularly those that exhibit **shift invariance**. By exploring this expanded gate set, we aim to identify gates that are practically implementable on current quantum hardware.

2 Overview

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1. Discuss the efficiency of the

3 Shift Invariance

3.1 Shift Invariance in QFT

To understand shift invariance, we begin with the QFT:

$$\text{QFT}|ch\rangle = \frac{1}{\sqrt{G}} \sum_g \rho(ch)|g\rangle \quad (1)$$

Here, c represents a coset label, and ρ is a specific representation. When applied to a superposition state $\sum_h |ch\rangle$, the QFT yields:

$$\text{QFT} \sum_{h \in H} |ch\rangle = \frac{\rho(c)}{\sqrt{G}} \sum_{hg} \rho(h)|g\rangle \quad (2)$$

Our goal is to eliminate the influence of the coset label c in the QFT, retaining only the information about the subgroup H .

To extract this information, we measure the quantum register. Removing the effect of c is equivalent to ensuring:

$$\left| \frac{\rho(c_1)}{\sqrt{G}} \sum_h \rho(h) \right| = \left| \frac{\rho(c_2)}{\sqrt{G}} \sum_h \rho(h) \right| \quad (3)$$

This equality holds because $\rho(c_1)$ and $\rho(c_2)$ differ only by a phase factor. While not a global phase, this ensures that the probability of measuring $|g\rangle$ remains the same.

4 Z Transform

We begin by analyzing the QFT and then explore how the classical Z transformation can be extended to its quantum counterpart.

4.1 Discrete Fourier Transform

A data vector can be expressed as:

$$\mathbf{x} = \sum x_i e_i \quad (4)$$

where e_i represents the natural basis:

$$e_i = [0, \dots, 0, 1, 0, \dots, 0] \quad (5)$$

(1 in the i -th position).

We perform a basis transformation $e_i \rightarrow f_i$, where:

$$f_k = \frac{1}{\sqrt{N}} \sum e^{2\pi i j \frac{k}{N}} e_j \quad (6)$$

This gives:

$$\mathbf{x} = \sum x_i e_i = \sum y_j f_j \quad (7)$$

and:

$$y_j = \frac{1}{\sqrt{N}} \sum e^{-2\pi i j \frac{k}{N}} x_k \quad (8)$$

We verify this transformation as follows:

$$x_l = e_l^T \sum y_k f_k = \frac{1}{N} \sum_{j,k} e^{-\frac{2\pi i j k}{N}} x_j e^{\frac{2\pi i l k}{N}} \quad (9.1)$$

$$= \frac{1}{N} \sum_j \sum_k e^{\frac{-2\pi i (j-l)k}{N}} x_j = x_l \quad (9.2)$$

If we use $|j\rangle$ to represent the basis e_j , then $e_j \rightarrow f_j$ could be written as

$$|k\rangle \rightarrow \frac{1}{\sqrt{N}} \sum e^{\frac{2\pi i j k}{N}} |j\rangle \quad (10)$$

Thus, from this aspect, we can see that the QFT is essentially a basis transformation. This establishes the quantum analog of the classical Fourier Transform.

4.2 Z transformation

We gain insights from the QFT when attempting to “quantize” a classical transformation. Specifically, we consider Z transformation as a kind of basis transformation. And analyze how the Z transformation interacts with the basis vectors e_i .

For the Z transformation, the operation $\mathcal{Z}$ acts on the basis e_k as follows:

$$\mathcal{Z} : e_k \rightarrow \sum_j j^{-k} e_j \quad (11)$$

Then, we have the quantum version of Z transformation:

$$|k\rangle \rightarrow \sum_j j^{-k} |j\rangle. \quad (12)$$

If Z transformation satisfy shift invariance?

$$\mathcal{Z} |c + H\rangle = \sum_h \sum_j j^{-(c+h)} |j\rangle = \sum_j j^{-c} \left(\sum_h j^{-h} \right) |j\rangle. \quad (13)$$

The probability to get j is with different coset is $|j^{-c_1} \sum j^{-h}|$ and $|j^{-c_2} \sum j^{-h}|$. It still carries the information of c . Do not satisfy shift invariance.

5 General Shift Invariance

5.1 Shift operation

For simple, we consider abelian group G first.

Let T be the operation that do the shift, and $|H\rangle = \sum |h\rangle$

$$T_c|g\rangle = |g + c\rangle, \forall g \in G. \quad (14)$$

Then the shift invariant condition could be expressed as

$$|\langle g | \text{QFT } T_c | H \rangle|^2 = |\langle g | \text{QFT } T_{c'} | H \rangle|^2, \quad \forall g, c \in G \quad (15)$$

5.2 Formal definition of Shift Invariance

Notice: I didn't find a 'formal' definition of shift invariance in the literature. However, we can define it as follows:

Definition 5.2.1 (Shift Invariance (over abelian group)): Given an abelian group G and any subgroup H , a unitary operation U is said to be shift invariant if

$$|\langle g | U | H \rangle|^2 = |\langle g | U T_c | H \rangle|^2, \quad \forall g, c \in G \quad (16)$$

5.3 Equivalent condition

$$|\langle g | U | H \rangle|^2 = |\langle g | U T_c | H \rangle|^2, \quad \forall g, c \in G \quad (17.1)$$

$$\Leftrightarrow \langle \langle g | \mathcal{U} | H \rangle \rangle = \langle \langle g | \mathcal{U} \mathcal{T}_c | H \rangle \rangle, \quad \forall g, c \in G \quad (17.2)$$

$$\Leftrightarrow \langle \langle g | \mathcal{U} | 0 \rangle \rangle = \langle \langle g | \mathcal{U} | k \rangle \rangle, \quad \forall k \in G \quad (17.3)$$

$$\Leftrightarrow \langle g | U | 0 \rangle \langle 0 | U^\dagger | g \rangle = \langle g | U | j \rangle \langle j | U^\dagger | g \rangle, \quad \forall j, g \in G \quad (17.4)$$

$$\Leftrightarrow |U_{0,g}|^2 = |U_{k,g}|^2, \quad \forall k, g \in G \quad (17.5)$$

We know that $\sum_i |U_{i,g}|^2 = 1$. Thus,

$$|U_{0,g}|^2 = |U_{k,g}|^2, \quad \forall k, g \in G \quad (18.1)$$

$$\Leftrightarrow |U_{i,g}|^2 = \frac{1}{\sqrt{N}} \quad (18.2)$$

Finally, we have

$$U \text{ satisfy shift invariance} \quad (19.1)$$

$$\Leftrightarrow U = \frac{1}{\sqrt{N}} \begin{pmatrix} e^{i\theta_{11}} & \dots & e^{i\theta_{1N}} \\ \vdots & & \vdots \\ e^{i\theta_{N1}} & \dots & e^{i\theta_{NN}} \end{pmatrix} \quad (19.2)$$

6 shift-invariance gate and Hadarmard-Ising gates

One example of shift-invariance gate is QFT, which includes $H^{\otimes n}$.

$$H^{\otimes n} = \frac{1}{\sqrt{N}} \begin{pmatrix} \pm 1 & \dots & \pm 1 \\ \vdots & & \vdots \\ \pm 1 & \dots & \pm 1 \end{pmatrix} \quad (20)$$

and Ising gates

$$\text{Is}(\boldsymbol{\theta}) = \exp\left(i \sum \theta_{ij} Z_i Z_j + i \sum \theta_i Z_i\right) \quad (21.1)$$

$$= \begin{pmatrix} e^{i\varphi_{11}} & & \\ & \ddots & \\ & & e^{i\varphi_{NN}} \end{pmatrix} \quad (21.2)$$

Hadamard-Ising gates $\text{Is}(\boldsymbol{\theta})H^{\otimes n}$ satisfy Equation 19, which means they satisfy the shift invariance.

7 What happen if we use Hadamard-Ising gates instead of QFT in Fourier sampling?

After measuring a quantum data table, we get $\sum_h |c + h\rangle$. Then, apply Hadamard-Ising gates

$$\text{Is}(\boldsymbol{\theta})H^{\otimes n} \sum_h |c + h\rangle = \sum_k e^{i \sum_{ij} k_i k_j \theta_{ij}} (-1)^{\mathbf{k} \cdot \mathbf{c}} |0\rangle^{\otimes(n-p)} |k\rangle \quad (22)$$

$$H^{\otimes n} |g_1 g_2 \dots g_n\rangle = \otimes(|0\rangle + (-1)^{g_j} |1\rangle) = \sum (-1)^{\sum k_j g_j} |k\rangle \quad (23)$$

$$H^{\otimes n} \sum_h |c + h\rangle = \sum_h \sum_k (-1)^{\sum k_j (c+h)_j} |k\rangle \quad (24)$$

Suppose $G \simeq \mathbb{Z}_{2^m}$, the element of H must be written by $q2^p$, where $p, q \in \mathbb{Z}^+$. It means that the binary format of h is

$$h = h_1 h_2 \dots h_{n-p} 0 \dots 0. \quad (25)$$

without loss of generality, let $c = 0..0c_{n-p+1} \dots c_n$. Thus,

$$h + c = h_1 h_2 \dots h_{n-p} c_{n-p+1} \dots c_n \quad (26)$$

Thus, $\sum_h (-1)^{\sum k_j (c+h)_j} \neq 0$ implies $k_1 = k_2 = \dots = k_{n-p} = 0$. For convience, denote $\mathbf{k} \cdot \mathbf{c} = k_{n-p+1} c_{n-p+1} + \dots + k_n c_n$. Then

$$H^{\otimes n} \sum_h |c + h\rangle = \sum_k (-1)^{\mathbf{k} \cdot \mathbf{c}} |0\rangle^{\otimes(n-p)} |k\rangle \quad (27)$$

And then apply Ising gates

$$\text{Is}(\boldsymbol{\theta})H^{\otimes n} \sum_h |c + h\rangle = \sum_k e^{i \sum_{ij} k_i k_j \theta_{ij}} (-1)^{\mathbf{k} \cdot \mathbf{c}} |0\rangle^{\otimes(n-p)} |k\rangle \quad (28)$$

7.1 Simple example

$$G \simeq \mathbb{Z}_4 = \{0, 1, 2, 3\} \quad (29.1)$$

$$G > H = \{0, 2\} = \{00, 10\} \quad (29.2)$$

Then, apply H:

$$H^{\otimes 2} |H\rangle = H^{\otimes 2} (|00\rangle + |10\rangle) \quad (30.1)$$

$$= (|00\rangle + |01\rangle + |10\rangle + |11\rangle) + (|00\rangle + |01\rangle - |10\rangle - |11\rangle) \quad (30.2)$$

$$= |00\rangle + |01\rangle \quad (30.3)$$

So after sampling, we know the first bit is always 0, which means $H = \{0, 2\}$

Compare to the QFT

$$\text{QFT}|H\rangle \quad (31.1)$$

$$= (|00\rangle + |01\rangle + |10\rangle + |11\rangle) + (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \quad (31.2)$$

$$= |00\rangle + |10\rangle \quad (31.3)$$

SURPRISE!!! If $G \simeq \otimes \mathbb{Z}_{2^{m_i}}$, it seems that the $H^{\otimes n}$ gate is enough!

7.2 Efficiency discussion

The algorithm of finding hidden subgroup with $\otimes \mathbb{Z}_{2^{m_i}}$ with Hadamard gates is:

1. apply $H^{\otimes n}$
2. measure the register m times
3. according to the measurement results, for each bit, decide the bit is always 0, or it is random (it's 0 or 1).

After measurement, denote the measurement result as $\{\mathbf{b}^{(i)}\}_{i=1}^m$, and $\mathbf{b}^{(i)} = b_1^{(i)} b_2^{(i)} \dots b_n^{(i)}$. For each bit b_j , we need to decide if it is always 0, or it is random (it's 0 or 1). The probability that we make wrong decision (the bit is random but we think it is always 0) is $(\frac{1}{2})^m$.

Thus,

$$\text{Prob}(\text{right decision}) \geq \left(1 - \frac{1}{2^m}\right)^n \geq 1 - \frac{n}{2^m} \geq 1 - \varepsilon \quad (32)$$

leads to

$$\frac{n}{2^m} \leq \varepsilon \quad (33.1)$$

$$m \geq \log_2 \left(\frac{n}{\varepsilon} \right) \quad (33.2)$$

We conclude that, when $m \geq \log_2 \left(\frac{n}{\varepsilon} \right)$, the probability of right decision is at least $1 - \varepsilon$.

7.3 Box 5.4 Factoring 15 quantum-mechanically (page 235 in Nielsen)

The equation (5.62) keep the same.

$$\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} |k\rangle |x^k \bmod N\rangle = \frac{1}{\sqrt{2^t}} [|0\rangle|1\rangle + |1\rangle|7\rangle + |2\rangle|4\rangle + \dots] \quad (34)$$

measure the second register, suppose we get $|4\rangle$. The first register is

$$|2\rangle + |6\rangle + |10\rangle + \dots \quad (35)$$

applying Hadamard gates, we have $\dots 00, \dots 01, \dots 10, \dots 11$. Thus, we get $r = 4, 7^4 \bmod 15 = 1$.

The rest part follows the box 5.4.

8 Advantage of QFT

From the above section, we found that for certain scenario, the Hadamard gates are enough for finding hidden subgroup. However, the QFT has it's own advantage: **When the given group G is not $\mathbb{Z}_2^{m_i}$, it can use a $\mathbb{Z}_2^{m_i}$ to approximate it.**

Let's illustrate this with an example.

Suppose $G \simeq \mathbb{Z}$, $G > H \simeq \mathbb{Z}/\mathbb{Z}_7$,

$$G = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}, \quad (36.1)$$

$$H = \{\dots - 14, -7, 0, 7, 14, \dots\} < G. \quad (36.2)$$

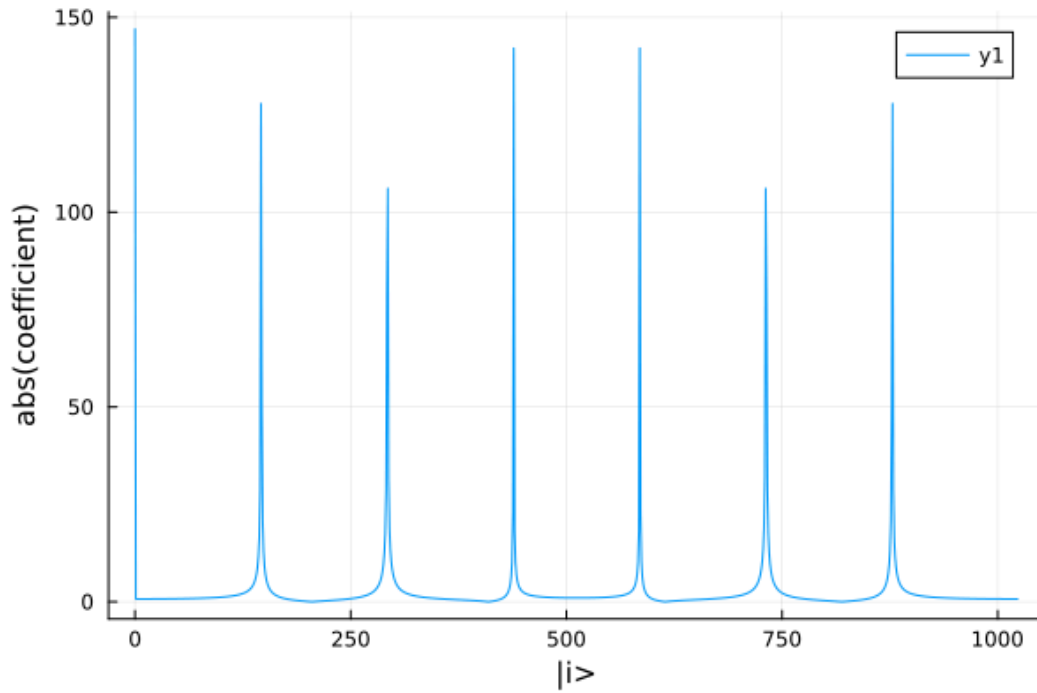
And a group function $f : G \rightarrow X$, with hidden subgroup H (we do not care what the function looks like, but we know that $f(x) = f(x + 7)$). Then, we can “**trancate**” the group G with a $\mathbb{Z}_{2^n}$, say $\mathbb{Z}_{2^{10}}$. And limit the function f to the truncated group $\mathbb{Z}_{2^{10}}$. Notice that the **trancated subgroup $H' = \{0, 7, \dots\}$ is not a subgroup of trancated group $\mathbb{Z}_{2^{10}}$.**

However, the QFT can still approximately find the hidden subgroup in this case!

$$\text{QFT}^{-1}|H'\rangle = \sum_k \left(\sum_{j=0}^{146} \exp\left(-\frac{2\pi i \cdot 7j \cdot k}{2^{10}}\right) \right) |k\rangle \quad (37)$$

Then, we print the unnormalized factor of the first 10 kets with largest probability:

$ 0\rangle : 147$	$ 146\rangle : 80 + 100i$	$ 147\rangle : -31 - 40i$	$ 292\rangle : -17 + 77i$	$ 293\rangle : 24 - 103i$
$ 439\rangle : 128 - 61i$	$ 585\rangle : 128 + 61i$	$ 731\rangle : 24 + 103i$	$ 732\rangle : -17 - 77i$	$ 878\rangle : 80 - 99i$



Apply continued fraction:

$$\frac{146}{1024} = \frac{1}{6 + \frac{1}{1+\dots}} \quad (38)$$

In the first iteration, we check that if $f(x + 6) = f(x)$. It fails. In the second iteration, $\frac{1}{6+\frac{1}{1}} = \frac{1}{7}$, we check that if $f(x + 7) = f(x)$. It success.

We can also try some other ket, like $|585\rangle$

$$\frac{585}{1024} = \frac{1}{1 + \frac{1}{1 + \frac{1}{3 + \frac{1}{146}}}} \quad (39)$$

when we check the fraction in third iteration, $\frac{1}{1 + \frac{1}{1 + \frac{1}{3}}} = \frac{4}{7}$, $f(x + 7) = f(x)$. It success.